PARAG PATIL

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SUMMARY

AI/ML Engineer who builds and ships production Generative AI and security-AI systems end to end. Acting AI Team Lead at Quasar Cybertech, where I developed the organisation's AI work across its products: an LLM-powered cyber threat-intelligence platform built solo, RAG and multi-agent AI assistant features, and an ML-driven SOC alert-triage module running on live Wazuh data. Skilled in LLM integration and token optimization, RAG, AI agents, machine learning, Python (FastAPI), PostgreSQL and React with TypeScript, with a Secure by Design focus across the SDLC.

EXPERIENCE

Quasar Cybertech **July 2025 - June 2026**

Trainee Developer *Nashik, India*

- Led all AI development for the organisation as its acting AI team lead, delivering the AI components of its flagship products (QPulse, QStellar) and setting core practices for Secure Coding by Design, data privacy and QA within the SDLC.
- Built LLM pipelines on Gemma-3-27B via the Google Gemini API, with token optimization (input trimming, batching, rate-limit control, de-duplication before model calls) to reduce API cost and latency.
- Evaluated multiple local LLMs via Ollama, deploying a 7B model (Qwen2-7B) for private on-premise inference, and moved to Google APIs as needs grew; fine-tuned a 3B-parameter model on an in-house dataset of cybersecurity reports.
- Built automated data-gathering modules and ETL pipelines that ingested and cleaned approximately 300,000 records for model training and semantic search.

PROJECTS

QPulse (https://qpulse.quasarcybertech.com/) **July 2025 - June 2026**

Cyber Threat-Intelligence Platform, Sole Developer

- Architected and delivered the entire platform end to end: React, TypeScript and Tailwind CSS frontend, FastAPI backend, and PostgreSQL with a standard ETL process from staging to production tables.
- Built an ingestion pipeline over global security feeds and Indian sources including CERT-In and RBI, where Gemma-3-27B turns each article into structured threat intelligence: severity, risk score, MITRE ATT&CK TTPs, CVEs, IOCs and recommendations.
- Added translation into 7 Indian regional languages, personalised email digests, and API hardening with rate limiting, reCAPTCHA and security headers.

QStellar (https://qstellar.co/) **July 2025 – June 2026**

Enterprise Cybersecurity Platform, AI Integration (5-member team)

- Built a document RAG pipeline with LangChain and ChromaDB: chunking strategies per file type, BGE embeddings, per-user collections, and cited answers from a local LLM cross-referenced against platform CVE data.
- Worked on the multi-agent AI assistant and built its specialist agents for assets, vulnerabilities, threats and executive summaries, with query routing, a scope guard and a grounding layer against hallucination.
- Built the QStellar web frontend in React and TypeScript, integrating it with the platform's AI modules.

QDetect, ML-Driven SOC Alert Triage (QStellar module) **Sole Developer**

- Built the module end to end: Wazuh SIEM integration (REST API and OpenSearch indexer) with scheduled sync of agents, alerts, vulnerabilities and CISA KEV data into PostgreSQL, served through 80+ FastAPI endpoints.
- Engineered a SOC suppression engine over 83,321 real alerts that removed about 84% as noise while never suppressing MITRE-mapped, critical, ransomware or endpoint signals.
- Developed a 4-model ML ensemble (Isolation Forest, DBSCAN, Random Forest, KNN) to classify the remaining grey-zone alerts, predict known threats and forecast trends from network events, plus an event-driven 0-100 security score per asset.

EDUCATION

K.K. Wagh Institute of Engineering Education & Research **2025**

B.E. in Artificial Intelligence & Data Science | *CGPA:* 8.24/10 | *Nashik, Maharashtra, India*

OTHER

- **Technical Skills:** Python (FastAPI) | Generative AI | LLM Integration | Token Optimization | LLM Fine-tuning | RAG (LangChain, ChromaDB) | AI Agents | Machine Learning (scikit-learn) | Ollama / Local LLMs | Google Gemini API | ETL Data Pipelines | PostgreSQL | React.js + TypeScript (Tailwind CSS) | Wazuh / SIEM | Secure Coding | Security by Design | SDLC | Docker | AWS | Nginx